

Assignment 1: Order of Operations

DO NOT WRITE ON THIS PAPER

Part A

Roll a die three times and use the rolls to replace A, B, and C in the following expression.

$$(A + 3)^B + 4 \times C - 10 + 5 - 35 \div 4^0$$

Example: If you rolled a 4, 5, and 1, your expression would be: $(4 + 3)^5 + 4 \times 3 - 10 + 5 - 35 \div 4^0$

Then, simplify the expression using the order of operations. You should have four clear sections of steps each corresponding to a letter in PEMDAS, organized into a table like the one shown below.

Letter	Name of Operation	Calculations
P		
E		
M D		
A S		

Part B

Griffin was tired and wrote the following expression and simplified it as follows:

Given Problem: $(3 - 2)^2 - (15 \div 3)$

Step 1: $= (3 - 2)^2 - 5$

Step 2: $= 3^2 - 2^2 - 5$

Step 3: $= 9 - 4 - 5$

Step 4: $= 5 - 5$

Answer: $= 0$

What mistake did Griffin make and at what step did it happen?

Correct any mistakes Griffin made and give the true value of the expression.

Part C

Variables are representations of numbers: they are place holders. Knowing this and order of operations are the following expressions equivalent? Explain your reasoning.

$$\text{Is } (x + 4)^2 \text{ equivalent to } x^2 + 16?$$

Hint: You can plug in any number for x and simplify. Try at least two different numbers to see if the expressions are really the same.

Question 1

Big Question: What is the Order of Operations? Why is it important that we follow them?

Use the example below to help support your answer as to why they are important.

$$5 + 9 \div 3$$

Additional Space